

DU Mathsoc Problem Solving: Set 2

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~Beginners Paper~

For any assistance or inquiries about the problem sets, please email me at: quizmaster.maths@csc.tcd.ie.

Problem 1. Let $p > 3$ be a prime number. Prove that $p^2 - 1$ is divisible by 24.

Problem 2. Prove that the incentre of any triangle is where all angle bisectors of that triangle intersect.

Problem 3. Pick 3 points at random on a circle of radius 1. What is the expected area of the triangle formed by those 3 points?

Problem 4. Prove that the Diophantine equation $x^3 + y^3 + z^3 = 32$ has no solution such that x, y , and z are all integers.

Problem 5. The perimeter of a rectangle is 20. What is the largest possible area that such a rectangle can have?